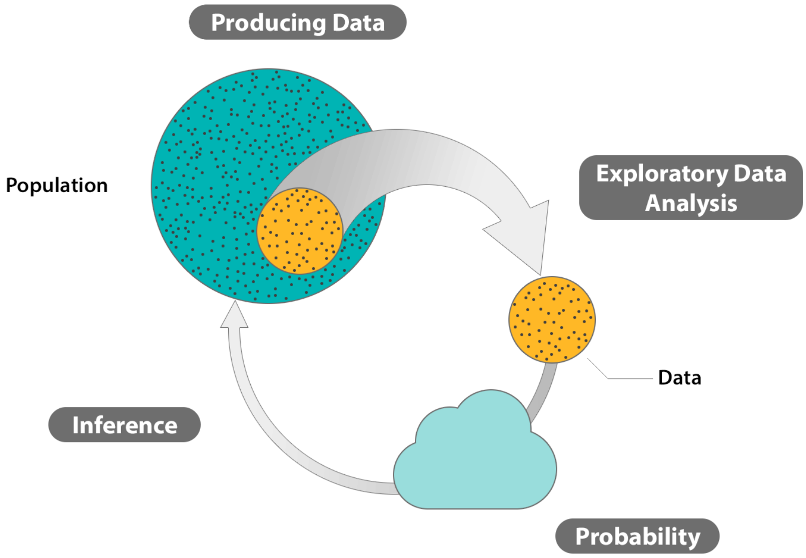
Contents

9 Module 9 | Introduction to Statistical Inference

9.1 Statistical Inference

Recall again the Big Picture, the four-step process that encompasses statistics: data production, exploratory data analysis, probability, and inference.

We are about to start the fourth part of the process and the final section of this course, where we draw on principles learned in the other units (exploratory data analysis, producing data, and probability) in order to accomplish what has been our ultimate goal all along: use a sample to infer (or draw conclusions) about the population from which it was drawn. As you'll see in the introduction, the specific form of inference called for depends on the type of variables involved: either a single categorical or quantitative variable or a combination of two variables whose relationship is of interest.



Learning Objectives

- Identify inference type, e.g., point estimation, interval estimation, and hypothesis testing.

9.1.1 Module 9 | Introduction to Inference (1 of 2)

The purpose of this introduction is to review how we got here and how the previous units fit together to allow us to make reliable inferences. Also, we will introduce the various forms of statistical inference that will be discussed in this section, and give a general outline of how this section is organized.

In the *Exploratory Data Analysis* unit, we learned to display and summarize data that were obtained from a sample. Regardless of whether we had one variable and we examined its distribution, or whether we had two variables and we examined the relationship between them, it was always understood that these summaries applied *only* to the data at hand; we did not attempt to make claims about the larger population from which the data were obtained.

Such generalizations were, however, a long-term goal from the very beginning of the course. For this reason, in the *Producing Data* unit (Sampling and Study Design), we took care to establish principles of sampling and study design that would be essential in order for us to claim that, to some extent, what is true for the sample should be also true for the larger population from which the sample originated. These principles should be kept in mind throughout this section on statistical inference, since the results that we will obtain will not hold if there was bias in the sampling process, or flaws in the study design under which variables' values were measured.

Perhaps the most important principle stressed in the Producing Data unit was that of randomization. Randomization is essential not only because it prevents bias but also because it permits us to rely on the laws of probability, which is the scientific study of random behavior.

In the *Probability* unit (Sampling Distributions), we established basic ideas of probability and the concept of random variables. We will focus on two *random variables*[1] of particular relevance: the sample mean ($\overline{X}$) and the sample proportion $\hat{p}$, and the last module was devoted to exploring their sampling distributions. We learned what probability theory tells us to expect from the values of the sample mean and the sample proportion, given that the corresponding population parameters – the population mean (μ) and the population proportion (p) – are known.

[1] Recall that a random variable is a variable whose values are numerical results of a random experiment.

As we mentioned in that module, the value of such results is more theoretical than practical, since in real-life situations we seldom know what is true for the entire population. All we know is what we see in the sample, and we want to use this information to say something concrete about the larger population. Probability theory has set the stage to accomplish this: learning what to expect from the value of sample mean, given that population mean takes a certain value, teaches us (as we'll soon learn) what to expect from the value of the unknown population mean, given that a particular value of sample mean has been observed. Similarly, since we have established how sample proportion behaves relative to population proportion, we will now be able to turn this around and say something about the value of population proportion, based on an observed sample proportion. This process – inferring something about the population based on what is measured in the sample – is (as you know) called *statistical inference*.

9.1.2 Module 9 | Introduction to Inference (2 of 2)

Learning Objectives

- Identify inference type, e.g., point estimation, interval estimation, and hypothesis testing.

We introduce three forms of statistical inference in this unit, each one representing a different way of using the information obtained in the sample to draw conclusions about the population. These forms are:

- Point estimation
- Interval estimation
- Hypothesis testing

Obviously, each one of these forms of inference will be discussed at length in this section, but it would be useful to get at least an intuitive sense of the nature of each of these inference forms, and the difference between them in terms of the type of conclusions they draw about the population based on the sample results.

In *point estimation*, we estimate an unknown parameter using a *single number* that is calculated from the sample data.

> ***Example*** **(A)**
>
> Based on sample results, we estimate that p, the proportion of all U.S. adults who are in favor of stricter gun control, is 0.6.

In *interval estimation*, we estimate an unknown parameter using an *interval of values* that is likely to contain the true value of that parameter (and state how confident we are that this interval indeed captures the true value of the parameter).

> ***Example*** **(B)**
>
> Based on sample results, we are 95% confident that p, the proportion of U.S. adults who are in favor of stricter gun control, is between 0.57 and 0.63.

In *hypothesis testing*, we have some claim about the population, and we check *whether or not the data* obtained from the sample *provide evidence against this claim.*

> ***Example*** **(C) 1**
>
> It was claimed that among all U.S. adults, about half are in favor of stricter gun control and about half are against it. In a recent poll of a random sample of 1,200 U.S. adults, 60% were in favor of stricter gun control. This data, therefore, provides some evidence against the claim.

> ***Example*** **(D) 2**
>
> It is claimed that among drivers 18-23 years of age (our population) there is no relationship between drunk driving and gender. A roadside survey collected data from a random sample of 5,000 drivers and recorded,their gender and whether they were drunk. The collected data showed roughly the same percent of drunk drivers among males

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and among females. These data, therefore, do not give us any reason to reject the claim that drunk driving is not related to gender.

10 Module 10 | Estimation

10.1 Inference For One Variable

As we mentioned at the end of the introduction, the first part of the inference unit will deal with inference for one variable. Recall that in the Exploratory Data Analysis (EDA) unit, when we learned about summarizing the data obtained from one variable (in the Examining Distributions module) we distinguished between two cases; categorical data and quantitative data.

We will make a similar distinction here in the inference unit. In the EDA unit, the type of variable determined the displays and numerical measures we used to summarize the data. In Inference, the type of variable of interest (categorical or quantitative) will determine what population parameter we are going to do inference for.

- When the variable of interest is *categorical*, the population parameter that we will infer about is the *population proportion (p)* associated with that variable. For example, if we are interested in studying opinions about the death penalty among U.S. adults, and thus our variable of interest is "death penalty (in favor/against)," we'll choose a sample of U.S. adults and use the collected data to make an inference about p – the proportion of U.S. adults who support the death penalty.
- When the variable of interest is *quantitative*, the population parameter that we infer about is the *population mean (μ)* associated with that variable. For example, if we are interested in studying the annual salaries in the population of teachers in a certain state, we'll choose a sample from that population and use the collected salary data to make an inference about μ, the mean annual salary of all teachers in that state.

Learning Objectives

- Determine point estimates in simple cases, and make the connection between the sampling distribution of a statistic, and its properties as a point estimator.
- Explain what a confidence interval represents and determine how changes in sample size and confidence level affect the precision of the confidence interval.
- Find confidence intervals for the population mean and the population proportion (when certain conditions are met), and perform sample size calculations.

10.2 Point Estimation

10.2.1 Module 10 | Point Estimation (1 of 2)

Learning Objectives

- Determine point estimates in simple cases, and make the connection between the sampling distribution of a statistic, and its properties as a point estimator.

Point estimation is the form of statistical inference in which, based on the sample data, we estimate the unknown parameter of interest using a *single* value (hence the name *point* estimation). As the following two examples illustrate, this form of inference is quite intuitive.

Example (E)

Suppose that we are interested in studying the IQ levels of students at Smart University (SU). In particular (since IQ level is a quantitative variable), we are interested in estimating μ, the mean IQ level of all the students at SU.

A random sample of 100 SU students was chosen, and their (sample) mean IQ level was found to be $\overline{x} = 115$

If we wanted to estimate μ, the population mean IQ level, by a single number based on the sample, it would make

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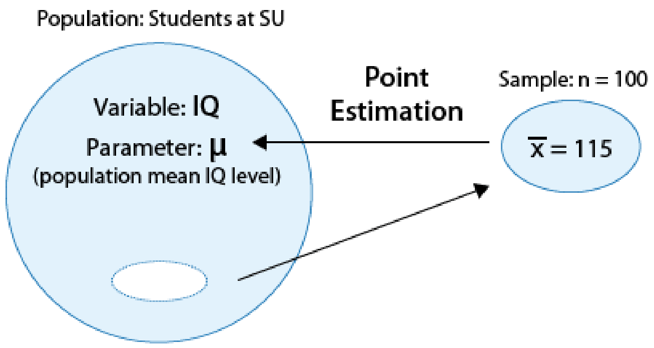
intuitive sense to use the corresponding quantity in the sample, the sample mean $\bar{x} = 115$. We say that 115 is the *point estimate* for μ, and in general, we'll always use $\bar{x}$ as the *point estimator* for μ. (Note that when we talk about the *specific value* (115), we use the term *estimate*, and when we talk in general about the *statistic* $\bar{x}$, we use the term *estimator*. The following figure summarizes this example:

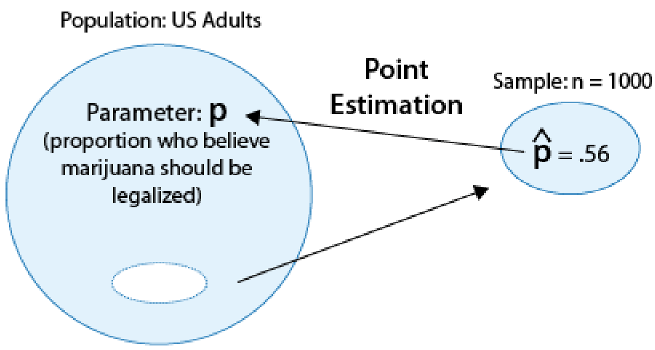


Here is another example.

Example (F)

Suppose that we are interested in the opinions of U.S. adults regarding legalizing the use of marijuana. In particular, we are interested in the parameter p, the proportion of U.S. adults who believe marijuana should be legalized.

Suppose a poll of 1,000 U.S. adults finds that 560 of them believe marijuana should be legalized. If we wanted to estimate p, the population proportion, using a single number based on the sample, it would make intuitive sense to use the corresponding quantity in the sample, the sample proportion $\hat{p} = \frac{560}{1000} = 0.56$. We say in this case that 0.56 is the *point estimate* for p, and in general, we'll always use $\hat{p}$ as the *point estimator* for p. (Note, again, that when we talk about the *specific value* (0.56), we use the term estimate, and when we talk in general about the *statistic* $\hat{p}$, we use the term *estimator.* Here is a visual summary of this example:



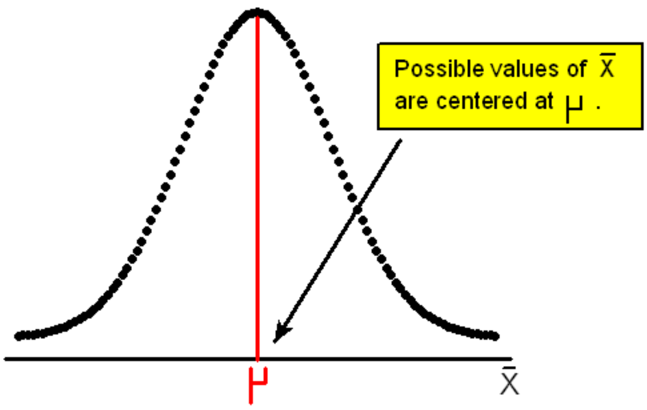
10.2.2 Module 10 | Point Estimation (2 of 2)

Learning Objectives

- Determine point estimates in simple cases, and make the connection between the sampling distribution of a statistic, and its properties as a point estimator.

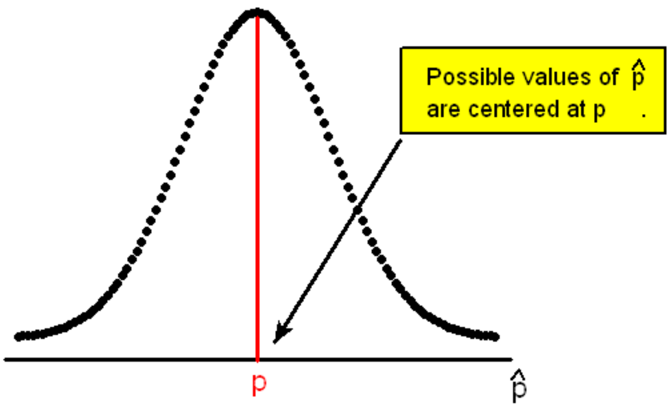
Comment 1 You may feel that since it is so intuitive, you could have figured out point estimation on your own, even without the benefit of an entire course in statistics. Certainly, our intuition tells us that the best estimator for should be $\bar{x}$, and the best estimator for p should be $\hat{p}$.

Probability theory does more than this; it actually gives an explanation (beyond intuition) *why* $\bar{x}$ and $\hat{p}$ are the good choices as point estimators for μ and p, respectively. In the Sampling Distributions module of the Probability unit, we learned about the sampling distributions of $\overline{X}$ and found that *as long as a sample is taken at random*, the distribution of sample means is exactly centered at the value of population mean.



$\overline{X}$ is therefore said to be an *unbiased estimator* for μ. Any particular sample mean might turn out to be less than the actual population mean, or it might turn out to be more. But in the long run, such sample means are "on target" in that they will not underestimate any more or less often than they overestimate.

Likewise, we learned that the sampling distribution of the sample proportion, $\hat{p}$, is centered at the population proportion p (as long as the sample is taken at random), thus making $\hat{p}$ an *unbiased estimator* for p.



As stated in the introduction, probability theory plays an essential role as we establish results for statistical inference. Our assertion above that sample mean and sample proportion are unbiased estimators is the first such instance.

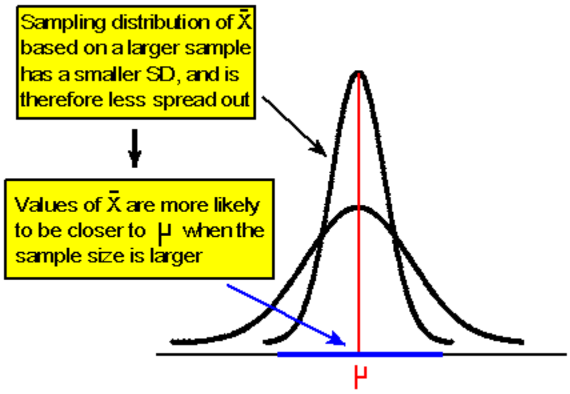
Comment 2 Notice how important the principles of sampling and design are for our above results: if the sample of U.S. adults in (example 2 on the previous page) was not random, but instead included predominantly college students, then 0.56 would be a biased estimate for p, the proportion of all U.S. adults who believe marijuana should be legalized. If the survey design were flawed, such as loading the question with a reminder about the dangers of marijuana leading to hard drugs, or a reminder about the benefits of marijuana for cancer patients, then 0.56 would be biased on the low or high side, respectively. Our point estimates are truly unbiased estimates for the population parameter only if the *sample is random and the study design is not flawed.*

Comment 3 Not only are sample mean and sample proportion on target as long as the samples are random, but their accuracy improves as sample size increases. Again, there are two "layers" here for explaining this.

Intuitively, larger sample sizes give us more information with which to pin down the true nature of the population. We can therefore expect the sample mean and sample proportion obtained from a larger sample to be closer to the population mean and proportion, respectively. In the extreme, when we sample the whole population (which is called a census), the sample mean and sample proportion will exactly coincide with the population mean and population proportion.

There is another layer here that, again, comes from what we learned about the sampling distributions of the sample mean and the sample proportion. Let's use the sample mean for the explanation.

Recall that the sampling distribution of the sample mean $\overline{X}$ is, as we mentioned before, centered at the population mean μ and has a standard deviation of $\frac{\sigma}{\sqrt{n}}$. As a result, as the sample size n increases, the sampling distribution of $\overline{X}$ gets less spread out. This means that values of $\overline{X}$ that are based on a larger sample are more likely to be closer to μ (as the figure below illustrates):



Similarly, since the sampling distribution of $\hat{p}$ is centered at p and has a standard deviation of $\sqrt{\frac{p(1-p)}{n}}$, which decreases as the sample size gets larger, values of $\hat{p}$ are more likely to be closer to p when the sample size is larger.

Let's Summarize We use $\hat{p}$ (sample proportion) as a point estimator for p (population proportion). It is an unbiased estimator: its long-run distribution is centered at p as long as the sample is random.

We use $\overline{x}$ (sample mean) as a point estimator for μ (population mean). It is an unbiased estimator: its long-run distribution is centered at μ as long as the sample is random.

In both cases, the larger the sample size, the more accurate the point estimator is. In other words, the larger the sample size, the more likely it is that the sample mean (proportion) is close to the unknown population mean (proportion).

10.3 Interval Estimation: Confidence Interval for the Population Mean

Learning Objectives

- Explain what a confidence interval represents and determine how changes in sample size and confidence level affect the precision of the confidence interval.
- Find confidence intervals for the population mean and the population proportion (when certain conditions are met), and perform sample size calculations.

Point estimation is simple and intuitive, but also a bit problematic. Here is why:

When we estimate, say, μ by the sample mean $\overline{x}$, we are almost guaranteed to make some kind of error. Even though we know that the values of $\overline{x}$ fall around μ, it is very unlikely that the value of $\overline{x}$ will fall exactly at μ.

Given that such errors are a fact of life for point estimates (by the mere fact that we are basing our estimate on one sample that is a small fraction of the population), these estimates are in themselves of limited usefulness, unless we are able to quantify the extent of the estimation error. Interval estimation addresses this issue. The idea behind *interval estimation* is, therefore, to enhance the simple point estimates by supplying information about the size of the error attached.

In this introduction, we'll provide examples that will give you a solid intuition about the basic idea behind interval estimation.

Example (G)

Consider the example that we discussed in the point estimation section:

Suppose that we are interested in studying the IQ levels of students in a Smart University (SU). In particular (since IQ level is a quantitative variable), we are interested in estimating μ, the mean IQ level of all the students in SU. A random sample of 100 SU students was chosen, and their (sample) mean IQ level was found to be $\overline{x} = 115$.

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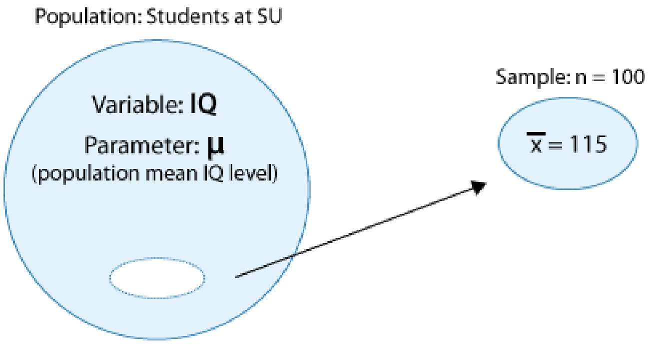


In point estimation we used $\overline{x} = 115$ as the point estimate for μ. However, we had no idea of what the estimation error involved in such an estimation might be. Interval estimation takes point estimation a step further and says something like:

> "I am 95% confident that by using the point estimate $\overline{x} = 115$ to estimate μ, I am off by no more than 3 IQ points. In other words, I am 95% confident that μ is within 3 of 115, or between 112 (= 115 − 3) and 118 (= 115 + 3)."

Yet another way to say the same thing is: I am 95% confident that μ is somewhere in (or covered by) the interval (112, 118).

Comment At this point you should not worry about, or try to figure out, how we got these numbers. We'll do that later. All we want to do here is make sure you understand the idea.)

Note that while point estimation provided just one number as an estimate for μ (115), interval estimation provides a whole interval of "plausible values" for μ (between 112 and 118), and also attaches the level of our confidence that this interval indeed includes the value of μ to our estimation (in our example, 95% confidence). The interval (112,118) is therefore called "a 95% confidence interval for μ."

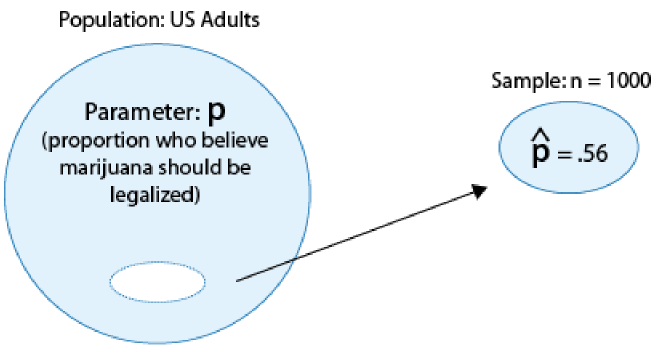
Let's look at another example:

Example (H)

Let's consider the second example from the point estimation section.

Suppose that we are interested in the opinions of U.S. adults regarding legalizing the use of marijuana. In particular, we are interested in the parameter p, the proportion of U.S. adults who believe marijuana should be legalized.

Suppose a poll of 1,000 U.S. adults finds that 560 of them believe marijuana should be legalized.



If we wanted to estimate p, the population proportion, by a single number based on the sample, it would make intuitive sense to use the corresponding quantity in the sample, the sample proportion $\hat{p} = \dfrac{560}{1000} = 0.56$.

Interval estimation would take this a step further and say something like:

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"I am 90% sure that by using 0.56 to estimate the true population proportion, p, I am off by (or, I have an error of) no more than 0.03 (or 3 percentage points). In other words, I am 90% confident that the actual value of p is somewhere between 0.53 (= 0.56 – 0.03) and 0.59 (= 0.56 + 0.03)."

Yet another way of saying this is: "I am 90% confident that p is covered by the interval (0.53, 0.59)."

In this example, (0.53, 0.59) is a 90% confidence interval for p.

Let's summarize The two examples showed us that the idea behind interval estimation is, instead of providing just one number for estimating an unknown parameter of interest, to provide an interval of plausible values of the parameter plus a level of confidence that the value of the parameter is covered by this interval.

We are now going to go into more detail and learn how these confidence intervals are created. As you'll see, the ideas that were developed in the "Sampling Distributions" module of the Probability unit will, again, be very important (as they were in point estimation).

We'll start by discussing confidence intervals for the population mean μ, and later discuss confidence intervals for the population proportion p.

10.3.1 Module 10 | Confidence Intervals for the Population Mean (1 of 8)

Overview As we mentioned in the introduction to interval estimation, we start by discussing interval estimation for the population mean μ. Here is a quick overview of how we introduce this topic.

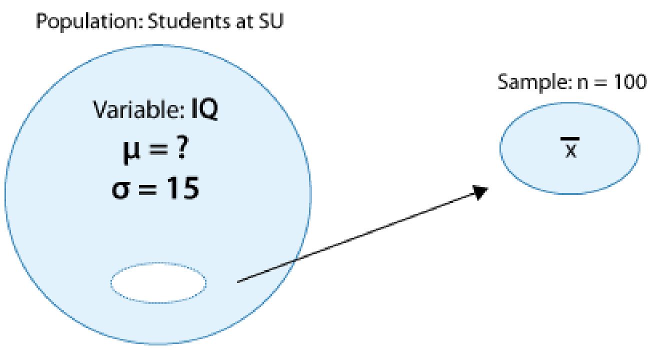
- Learn how a 95% confidence interval for the population mean μ is constructed and interpreted.
- Generalize to confidence intervals with other levels of confidence (for example, what if we want a 99% confidence interval?).
- Understand more broadly the structure of a confidence interval and the importance of the margin of error.
- Understand how the precision of interval estimation is affected by the confidence level and sample size.
- Learn under which conditions we can safely use the methods that are introduced in this section.

Recall the IQ example:

***Example* (I)**

Suppose that we are interested in studying the IQ levels of students at Smart University (SU). In particular (since IQ level is a quantitative variable), we are interested in estimating μ, the mean IQ level of all the students at SU.

We will assume that from past research on IQ scores in different universities, it is known that the IQ standard deviation in such populations is σ = 15. In order to estimate μ, a random sample of 100 SU students was chosen, and their (sample) mean IQ level is calculated (let's not assume, for now, that the value of this sample mean is 115, as before).



We will now show the rationale behind constructing a 95% confidence interval for the population mean μ.

- We learned in the "Sampling Distributions" module of probability that according to the central limit theorem, the sampling distribution of the sample mean $\overline{X}$ is approximately normal with a mean of μ and

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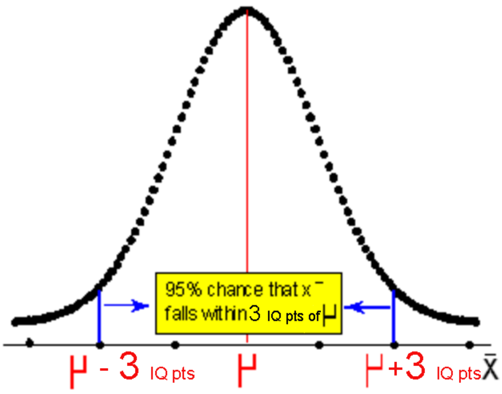
standard deviation of $\frac{\sigma}{\sqrt{n}}$. In our example, then, (where $\sigma = 15$ and $n = 100$), the possible values of $\overline{X}$, the sample mean IQ level of 100 randomly chosen students, is approximately normal, with mean μ and standard deviation $\frac{15}{\sqrt{100}} = 1.5$.

- Next, we recall and apply the Standard Deviation Rule for the normal distribution, and in particular its second part:

  There is a 95% chance that the sample mean we get in our sample falls within $2 \times 1.5 = 3$ of μ.



- Obviously, if there is a certain distance between the sample mean and the population mean, we can describe that distance by starting at either value. So, if the sample mean ($\overline{x}$) falls within a certain distance of the population mean μ, then the population mean μ falls within the same distance of the sample mean.

Therefore, the statement, "There is a 95% *chance* that the *sample* mean $\overline{x}$ falls within 3 units of μ" can be rephrased as: "We are 95% *confident* that the *population* mean μ falls within 3 units of $\overline{x}$."

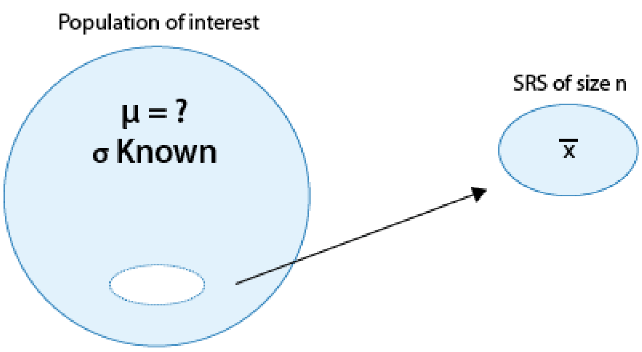
So, if we happen to get a sample mean of $\overline{x} = 115$, then we are 95% sure that μ falls within 3 of 115, or in other words that μ is covered by the interval $(115 - 3, 115 + 3) = (112, 118)$.

(On later pages, we will use similar reasoning to develop a general formula for a confidence interval.)

Comment Note that the first phrasing is about $\overline{x}$, which is a random variable; that's why it makes sense to use probability language. But the second phrasing is about μ, which is a parameter, and thus is a "fixed" value that doesn't change, and that's why we shouldn't use probability language to discuss it. This point will become clearer after you do the activities on the next page.

10.3.2 Module 10 | Confidence Intervals for the Population Mean (2 of 8)

The General Case Let's generalize the IQ example. Suppose that we are interested in estimating the unknown population mean (μ) based on a random sample of size n. Further, we assume that the population standard deviation (σ) is known.



The values of $\bar{x}$ follow a normal distribution with (unknown) mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$ (known, since both σ and n are known). By the (second part of the) Standard Deviation Rule, this means that:

There is a 95% chance that our sample mean ($\bar{x}$) will fall within $2\frac{\sigma}{\sqrt{n}}$ of μ,

which means that:

We are 95% confident that μ falls within $2\frac{\sigma}{\sqrt{n}}$ of our sample mean ($\bar{x}$).

Or, in other words, a 95% confidence interval for the population mean μ is:

$$\left(\bar{x} - 2\frac{\sigma}{\sqrt{n}}, \bar{x} + 2\frac{\sigma}{\sqrt{n}}\right).$$

Here, then, is the *general result:*

Suppose a random sample of size n is taken from a normal population of values for a quantitative variable whose mean (μ) is unknown, when the standard deviation (σ) is given. A 95% confidence interval (CI) for μ is: $\bar{x} \pm 2\frac{\sigma}{\sqrt{n}}$.

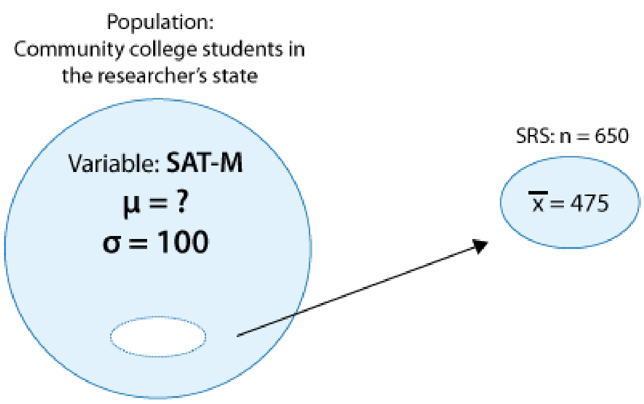
Comment Note that for now we require the population standard deviation (σ) to be known. Practically, σ is rarely known, but for some cases, especially when a lot of research has been done on the quantitative variable whose mean we are estimating (such as IQ, height, weight, scores on standardized tests), it is reasonable to assume that σ is known. Eventually, we will see how to proceed when σ is unknown, and must be estimated with sample standard deviation (s).

Let's look at another example.

Example (J)

An educational researcher was interested in estimating μ, the mean score on the math part of the SAT (SAT-M) of all community college students in his state. To this end, the researcher has chosen a random sample of 650 community college students from his state, and found that their average SAT-M score is 475. Based on a large body of research that was done on the SAT, it is known that the scores roughly follow a normal distribution with the standard deviation $\sigma = 100$.

Here is a visual representation of this story, which summarizes the information provided:



Based on this information, let's estimate μ with a 95% confidence interval.

Using the formula we developed before, $\bar{x} \pm 2\frac{\sigma}{\sqrt{n}}$, a 95% confidence interval for μ is:

$$\left(475 - 2\frac{100}{\sqrt{650}}, 475 + 2\frac{100}{\sqrt{650}}\right),$$

which is $(475 - 7.8, 475 + 7.8) = (467.2, 482.8)$. In this case, it makes sense to round, since SAT scores can be only whole numbers, and say that the 95% confidence interval is (467, 483).

We are not done yet. An equally important part is to *interpret what this means in the context of the problem.*

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We are 95% confident that the mean SAT-M score of all community college students in the researcher's state is covered by the interval (467, 483). Note that the confidence interval was obtained by taking 475 ± 8 (rounded). This means that we are 95% confident that by using the sample mean ($\bar{x}$ = 475) to estimate μ, our error is no more than 8.

We just saw that one interpretation of a 95% confidence interval is that we are 95% confident that the population mean (μ) is contained in the interval. Another useful interpretation in practice is that, given the data, the confidence interval represents the set of plausible values for the population mean μ.

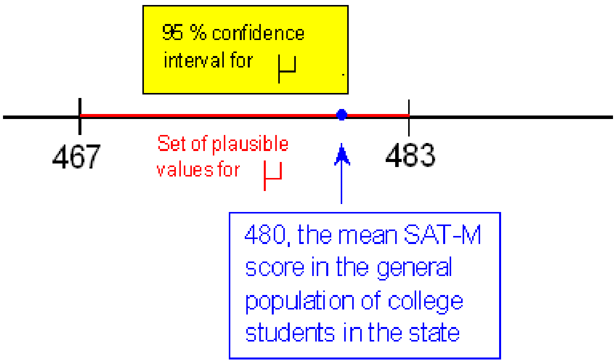
Example (K)

As an illustration, let's return to the example of mean SAT-Math score of community college students. Recall that we had constructed the confidence interval (467, 483) for the unknown mean SAT-M score for all community college students.

Here is a way that we can use the confidence interval:

Do the results of this study provide evidence that μ, the mean SAT-M score of community college students, is lower than the mean SAT-M score in the general population of college students in that state (which is 480)?

The 95% confidence interval for μ was found to be (467, 483). Note that 480, the mean SAT-M score in the general population of college students in that state, falls inside the interval, which means that it is one of the plausible values for μ.



This means that μ could be 480 (or even higher, up to 483), and therefore we cannot conclude that the mean SAT-M score among community college students in the state is lower than the mean in the general population of college students in that state. (Note that the fact that most of the plausible values for μ fall below 480 is not a consideration here.)

Comment Recall that in the formula for the 95% confidence interval for μ, $\bar{x} \pm 2\frac{\sigma}{\sqrt{n}}$, the 2 comes from the Standard Deviation Rule, which says that any normal random variable (in our case $\overline{X}$), has a 95% chance (or probability of 0.95) of taking a value that is within 2 standard deviations of its mean.

As you recall from the discussion about the normal random variable, this is only an approximation, and to be more accurate, there is a 95% chance that a normal random variable will take a value within 1.96 standard deviations of its mean. Therefore, a more accurate formula for the 95% confidence interval for μ is $\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$, which you'll find in most introductory statistics books. In this course, we'll use 2 (and not 1.96), which is close enough for our purposes.

10.3.3 Module 10 | Confidence Intervals for the Population Mean (3 of 8)

Other Levels of Confidence The most commonly used level of confidence is 95%. However, we may wish to increase our level of confidence and produce an interval that is almost certain to contain μ. Specifically, we may want to report an interval for which we are 99% confident – rather than only 95% confident – that it contains the unknown population mean.

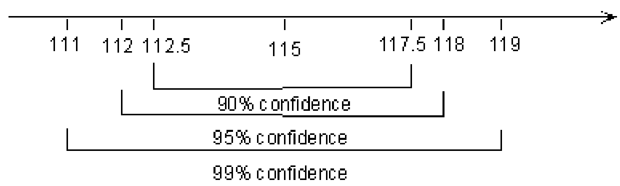
Using the same reasoning as in the last comment, in order to create a 99% confidence interval for μ, we should ask:

There is a probability of 0.99 that any normal random variable takes values within how many standard deviations of its mean? The precise answer is 2.576, and therefore, a 99% confidence interval for μ is $\overline{x} \pm 2.576\frac{\sigma}{\sqrt{n}}$.

Another commonly used level of confidence is a 90% level of confidence. Since there is a probability of 0.90 that any normal random variable takes values within 1.645 standard deviations of its mean, the 90% confidence interval for μ is $\overline{x} \pm 1.645\frac{\sigma}{\sqrt{n}}$.

> ***Example*** (L)
>
> Let's go back to our first example, the IQ example:
>
> The IQ level of students at a particular university has an unknown mean, μ, and a known standard deviation, $\sigma = 15$. A simple random sample of 100 students is found to have a sample mean IQ, $\overline{x} = 115$. Estimate μ with 90%, 95%, and 99% confidence intervals.
>
> A 90% confidence interval for μ is $\overline{x} \pm 1.645\frac{\sigma}{\sqrt{n}} = 115 \pm 1.645\left(\frac{15}{\sqrt{100}}\right) = 115 \pm 2.5 = (112.5,\ 117.5)$.
>
> A 95% confidence interval for μ is $\overline{x} \pm 2\frac{\sigma}{\sqrt{n}} = 115 \pm 2\left(\frac{15}{\sqrt{100}}\right) = 115 \pm 3.0 = (112,\ 118)$.
>
> A 99% confidence interval for μ is $\overline{x} \pm 2.576\frac{\sigma}{\sqrt{n}} = 115 \pm 2.576\left(\frac{15}{\sqrt{100}}\right) = 115 \pm 4.0 = (111,\ 119)$.

Note that the more confidence I require, the wider the confidence interval for μ (pronounced and sometimes noted as "mu"). The 99% confidence interval is wider than the 95% confidence interval, which is wider than the 90% confidence interval.



Confidence Intervals for mu

This is not very surprising, given that in the 99% interval we multiply the standard deviation by 2.576, in the 95% by 2, and in the 90% only by 1.645. Beyond this numerical explanation, there is a very clear intuitive explanation and an important implication of this result.

Let's start with the intuitive explanation. The more certain I want to be that the interval contains the value of μ, the more plausible values the interval needs to include in order to account for that extra certainty. I am 95% certain that the value of μ is one of the values in the interval (112,118). In order to be 99% certain that one of the values in the interval is the value of μ, I need to include more values, and thus provide a wider confidence interval.

In our example, the *wider* 99% confidence interval (111, 119) gives us a *less precise* estimation about the value of μ than the narrower 90% confidence interval (112.5, 117.5), because the smaller interval "narrows in" on the plausible values of μ.

The important practical implication here is that researchers must decide whether they prefer to state their results with a higher level of confidence or produce a more precise interval. In other words,

There is a trade-off between the level of confidence and the precision with which the parameter is estimated.

The price we have to pay for a higher level of confidence is that the unknown population mean will be estimated with less precision (i.e., with a wider confidence interval). If we would like to estimate μ with more precision (i.e., a narrower confidence interval), we will need to sacrifice and report an interval with a lower level of confidence.

10.3.4 Module 10 | Confidence Intervals for the Population Mean (4 of 8)

So far, we've developed the confidence interval for the population mean from scratch, based on results from probability, and discussed the trade-off between the level of confidence and the precision of the interval. The price you pay for a higher level of confidence is a lower level of precision of the interval (i.e., a wider interval).

Is there a way to bypass this trade-off? In other words, is there a way to increase the precision of the interval (i.e., make it narrower) *without* compromising on the level of confidence? We will answer this question shortly, but first we need to get a deeper understanding of the different components of the confidence interval and its structure.

Understanding the general structure of the confidence intervals We explored the confidence interval for μ for different levels of confidence and found that, in general, it has the following form:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}},$$

where z^* is a general notation for the multiplier that depends on the level of confidence. As we discussed before:

For a 90% level of confidence, $z^* = 1.645$

For a 95% level of confidence, $z^* = 2$ (or 1.96 if you want to be really precise)

For a 99% level of confidence, $z^* = 2.576$

To start our discussion about the structure of the confidence interval, let's denote the $z^* \frac{\sigma}{\sqrt{n}}$ formula by m.

The confidence interval, then, has the form: $\bar{x} \pm m$:

$$\boxed{\bar{x}} \pm \underbrace{\boxed{z^* \cdot \frac{\sigma}{\sqrt{n}}}}_{m}$$

$\bar{x}$ is the sample mean, the point estimator for the unknown population mean (μ).

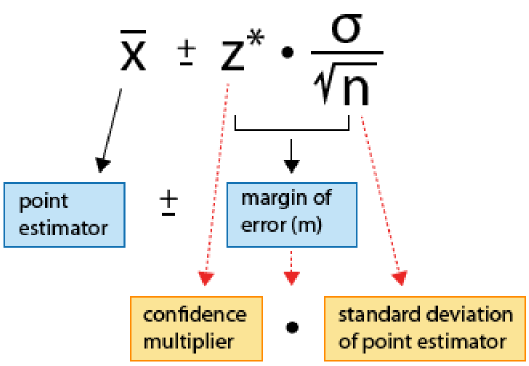
m is called the **margin of error**, since it represents the maximum estimation error for a given level of confidence.

For example, for a 95% confidence interval, we are 95% sure that our estimate will not depart from the true population mean by more than m, the margin of error.

m is further made up of the product of two components:

- z^*, the **confidence multiplier**, and
- $\frac{\sigma}{\sqrt{n}}$, which is the standard deviation of $\overline{X}$, the point estimator of μ.

Here is a summary of the different components of the confidence interval and its structure:



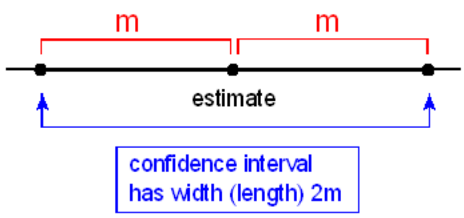
This structure:

$$\text{estimate} \pm \text{margin of error},$$

where the margin of error is further composed of the product of a confidence multiplier and the standard deviation (or, as we'll see, the standard error) is the general structure of all confidence intervals that we will encounter in this course.

Obviously, even though each confidence interval has the same components, what these components actually are is different from confidence interval to confidence interval, depending on what unknown parameter the confidence interval aims to estimate.

Since the structure of the confidence interval is such that it has a margin of error on either side of the estimate, it is centered at the estimate (in our case, $\overline{x}$), and its width (or length) is exactly twice the margin of error:



The margin of error, m, is therefore "in charge" of the width (or precision) of the confidence interval, and the estimate is in charge of its location (and has no effect on the width).

10.3.5 Module 10 | Confidence Intervals for the Population Mean (5 of 8)

Let us now go back to the confidence interval for the mean, and more specifically, to the question that we posed at the beginning of the previous page:

Is there a way to increase the precision of the confidence interval (i.e., make it narrower) *without* compromising on the level of confidence?

Since the width of the confidence interval is a function of its margin of error (given by $z^* \frac{\sigma}{\sqrt{n}}$), let's look closely at the margin of error of the confidence interval for the mean and see how it can be reduced. Since z^* controls the level of confidence, we can rephrase our question above in the following way:

Is there a way to reduce this margin of error other than by reducing z^*?

If you look closely at the margin of error, you'll see that the answer is yes. We can do that by increasing the sample size n (since it appears in the denominator).

Let's look at an example first and then explain why increasing the sample size is a way to increase the precision of the confidence interval *without* compromising on the level of confidence.

Example (M)

Recall the IQ example:

The IQ level of students at a particular university has an unknown mean (μ) and a known standard deviation of $\sigma = 15$. A simple random sample of 100 students is found to have the sample mean IQ $\overline{x} = 115$. A 95% confidence interval for μ in this case is:

$$\overline{x} \pm 2\frac{\sigma}{\sqrt{n}} = 115 \pm 2\left(\frac{15}{\sqrt{100}}\right) = 115 \pm 3.0 = (112,\ 118)$$

Note that the margin of error is $m = 3$, and therefore the width of the confidence interval is 6.

Now, what if we change the problem slightly by increasing the sample size, and assume that it was 400 instead of 100?

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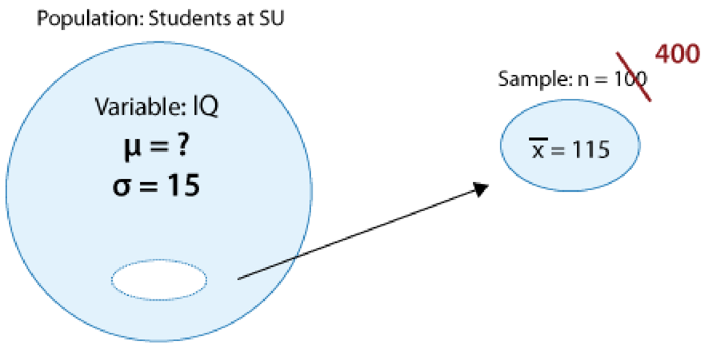


In this case, the 95% confidence interval for μ is:

$$\bar{x} \pm 2\frac{\sigma}{\sqrt{n}} = 115 \pm 2\frac{15}{\sqrt{400}} = 115 \pm 1.5 = (113.5,\ 116.5)$$

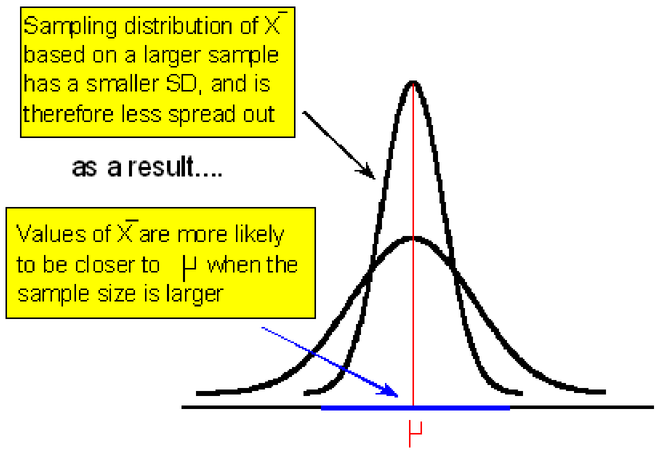
The margin of error here is only $m = 1.5$, and thus the width is only 3.

Note that for the same level of confidence (95%) we now have a narrower, and thus more precise, confidence interval.

Let's try to understand why a larger sample size will reduce the margin of error for a fixed level of confidence. There are three ways to explain it: mathematically, using probability theory, and intuitively.

We've already alluded to the mathematical explanation; the margin of error is $z^* \frac{\sigma}{\sqrt{n}}$, and since n, the sample size, appears in the denominator, increasing n will reduce the margin of error.

As we saw in our discussion about point estimates, probability theory tells us that:



This explains why with a larger sample size the margin of error (which represents how far apart we believe $\bar{x}$ might be from μ for a given level of confidence) is smaller.

On an intuitive level, if our estimate $\bar{x}$ is based on a larger sample (i.e., a larger fraction of the population), we have more faith in it, or it is more reliable, and therefore we need to account for less error around it.

Comment While it is true that for a given level of confidence, increasing the sample size increases the precision of our interval estimation, in practice, increasing the sample size is not always possible. Consider a study in which there is a non-negligible cost involved for collecting data from each participant (an expensive medical procedure, for example). If the study has some budgetary constraints, which is usually the case, increasing the sample size from 100 to 400 is just not possible in terms of cost-effectiveness. Another instance in which increasing the sample size is impossible is when a larger sample is simply not available, even if we had the money to afford it. For example, consider a study on the effectiveness of a drug on curing a very rare disease among children. Since the disease is rare, there are a limited number of children who could be participants. This is the reality of statistics. Sometimes theory collides with reality, and you just do the best you can.

Sample Size Calculations As we just learned, for a given level of confidence, the sample size determines the size of the margin of error and thus the width, or precision, of our interval estimation. This process can be reversed.

In situations where a researcher has some flexibility as to the sample size, the researcher can calculate in advance what the sample size is that he/she needs in order to be able to report a confidence interval with a certain level of confidence and a certain margin of error. Let's look at an example.

Example **(N)**

Recall the example about the SAT-M scores of community college students.

An educational researcher is interested in estimating μ, the mean score on the math part of the SAT (SAT-M) of all community college students in his state. To this end, the researcher has chosen a random sample of 650 community college students from his state, and found that their average SAT-M score is 475. Based on a large body of research that was done on the SAT, it is known that the scores roughly follow a normal distribution, with the standard deviation $\sigma = 100$.

The 95% confidence interval for μ is $(475 - 2\frac{100}{\sqrt{650}},\ 475 + 2\frac{100}{\sqrt{650}})$, which is roughly 475 ± 8, or $(467, 484)$. For a sample size of $n = 650$, our margin of error is 8.

Now, let's think about this problem in a slightly different way:

An educational researcher is interested in estimating μ, the mean score on the math part of the SAT (SAT-M) of all community college students in his state with a margin of error of (only) 5, at the 95% confidence level. What is the sample size needed to achieve this? (σ, of course, is still assumed to be 100).

To solve this, we set: $m = 2 \cdot \frac{100}{\sqrt{n}} = 5$ so $\sqrt{n} = \frac{2(100)}{5}$ and

$$n = \left(\frac{2(100)}{5}\right)^2 = 1600$$

So, for a sample size of 1,600 community college students, the researcher will be able to estimate μ with a margin of error of 5, at the 95% level. In this example, we can also imagine that the researcher has some flexibility in choosing the sample size, since there is a minimal cost (if any) involved in recording students' SAT-M scores, and there are many more than 1,600 community college students in each state.

Rather than take the same steps to isolate n every time we solve such a problem, we may obtain a general expression for the required n for a desired margin of error m and a certain level of confidence.

Since $m = z\frac{\sigma}{\sqrt{n}}$ is the formula to determine m for a given n, we can use simple algebra to express n in terms of m (multiply both sides by the square root of n, divide both sides by m, and square both sides) to get

$$n = \left(\frac{z^*\sigma}{m}\right)^2.$$

Comment Clearly, the sample size n must be an integer. In the previous example we got $n = 1,600$, but in other situations, the calculation may give us a non-integer result. In these cases, we should always *round up to the next highest integer.*

Using this "conservative approach," we'll achieve an interval at least as narrow as the one desired.

Example **(O)**

IQ scores are known to vary normally with a standard deviation of 15. How many students should be sampled if we want to estimate the population mean IQ at 99% confidence with a margin of error equal to 2?

$$n = \left(\frac{z^*\sigma}{m}\right)^2 = \left(\frac{2.576(15)}{2}\right)^2 = 373.26$$

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Round up to be safe, and take a sample of 374 students.

Comment In the preceding activity, you saw that in order to calculate the sample size when planning a study, you needed to know the population standard deviation, sigma (σ). In practice, sigma is usually not known, because it is a parameter. (The rare exceptions are certain variables like IQ score or standardized tests that might be constructed to have a particular known sigma.)

Therefore, when researchers wish to compute the required sample size in preparation for a study, they use an *estimate* of sigma. Usually, sigma is estimated based on the standard deviation obtained in prior studies.

10.3.7 Module 10 | Confidence Intervals for the Population Mean (7 of 8)

We are almost done with this section. We need to discuss just a few more questions:

- Is it always okay to use the confidence interval we developed for μ when σ is known?
- What if σ is unknown?
- How can we use statistical software to calculate confidence intervals for us?

When Is It Safe to Use the Confidence Interval We Developed? One of the most important things to learn with any inference method is the conditions under which it is safe to use it. It is very tempting to apply a certain method, but if the conditions under which this method was developed are not met, then using this method will lead to unreliable results, which can then lead to wrong and/or misleading conclusions. As you'll see throughout this section, we always discuss the conditions under which each method can be safely used.

In particular, the confidence interval for μ (when σ is known), $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$, was developed assuming that the sampling distribution of $\bar{X}$ is normal; in other words, that the Central Limit Theorem applies. In particular, this allowed us to determine the values of z^*, the confidence multiplier, for different levels of confidence.

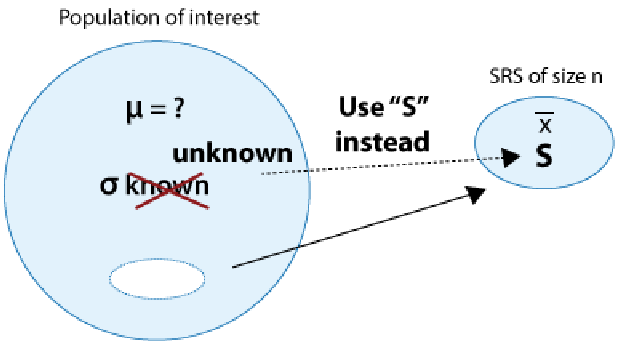
First, *the sample must be random.* Assuming that the sample is random, recall from the Probability unit that the Central Limit Theorem works when the *sample size is large* (a common rule of thumb for "large" is $n > 30$), or, for *smaller sample sizes*, if it is known that the quantitative *variable* of interest is *distributed normally* in the population. The only situation in which we cannot use the confidence interval, then, is when the sample size is small and the variable of interest is not known to have a normal distribution. In that case, other methods, called nonparametric methods, which are beyond the scope of this course, need to be used. This can be summarized in the following table:

	Small sample size	Large sample size
Variable varies normally	✓	✓
Variable doesn't vary normally	✗	✓

What if σ is unknown? [Optional Reading] As we discussed earlier, when variables have been well-researched in different populations it is reasonable to assume that the population standard deviation (σ) is known. However, this is rarely the case. What if σ is unknown?

Well, there is some good news and some bad news.

The good news is that we can easily replace the population standard deviation, σ, with the *sample* standard deviation, s.



The bad news is that once σ has been replaced by s, we lose the Central Limit Theorem, together with the normality of $\overline{X}$, and therefore the confidence multipliers z^* for the different levels of confidence $(1.645, 2, 2.576)$ are (generally) not accurate any more. The new multipliers come from a different distribution called the "t distribution" and are therefore denoted by t^* (instead of z^*).

The confidence interval for the population mean (μ) when (σ) is unknown is therefore:

$$\overline{x} \pm t^* \frac{s}{\sqrt{n}}$$

(Note that this interval is very similar to the one when σ is known, with the obvious changes: s replaces σ, and t^* replaces z^* as discussed above.)

There is an important difference between the confidence multipliers we have used so far (z^*) and those needed for the case when σ is unknown (t^*). Unlike the confidence multipliers we have used so far (z^*), which depend only on the level of confidence, the new multipliers (t^*) have the *added complexity* that they *depend on both the level of confidence and on the sample size* (for example, the t^* used in a 95% confidence when $n = 10$ is different from the t^* used when $n = 40$). Due to this added complexity in determining the appropriate t^*, researchers rely heavily on software in this case.

Comments

1. Since it is quite rare that σ is known, this interval (sometimes called a *one-sample t confidence interval*) is more commonly used as the confidence interval for estimating μ. (Nevertheless, we could not have presented it without our extended discussion up to this point, which also provided you with a solid understanding of confidence intervals.)

2. The quantity $\frac{s}{\sqrt{n}}$ is called the *standard error* of $\overline{X}$. The central limit theorem tells us that $\frac{\sigma}{\sqrt{n}}$ is the *standard deviation* of $\overline{X}$ (and this is the quantity used in confidence interval when σ is known). In general, whenever we replace parameters with their sample counterparts in the standard deviation of a statistic, the resulting quantity is called the standard error of the statistic. In this case, we replaced σ with its sample counterpart (s), and thus $\frac{s}{\sqrt{n}}$ is the *standard error* of (the statistic) $\overline{X}$.

3. As before, to safely use this confidence interval, the sample *must be random*, and the only case when this interval cannot be used is when the sample size is small and the variable is not known to vary normally.

Final comment It turns out that for large values of n, the t^* multipliers are not that different from the z^* multipliers, and therefore using the interval formula:

$$\overline{x} \pm z^* \frac{s}{\sqrt{n}}$$

for μ when σ is unknown provides a pretty good approximation.

10.3.8 Module 10 | Confidence Intervals for the Population Mean (8 of 8)

Let's summarize

- When the population is normal and/or the sample is large, a confidence interval for unknown population mean μ when σ is known is:

  $$\overline{x} \pm z\frac{\sigma}{\sqrt{n}},$$

  where z^* is 1.645 for 90% confidence, 2 for 95% confidence, and 2.576 for 99% confidence.

- There is a trade-off between the level of confidence and the precision of the interval estimation. The price we have to pay for more precision is sacrificing level of confidence.

- The general form of confidence intervals is an estimate ± the margin of error (m). In this case, the estimate = $\overline{x}$ and $m = z\frac{\sigma}{\sqrt{n}}$. The confidence interval is therefore centered at the estimate and its width is exactly $2m$.

- For a given level of confidence, the width of the interval depends on the sample size. We can therefore do a sample size calculation to figure out what sample size is needed in order to get a confidence interval with a desired margin of error m, and a certain level of confidence (assuming we have some flexibility with the sample size). To do the sample size calculation we use:

  $$n = \left(\frac{z^*\sigma}{m}\right)^2$$

  (and round *up* to the next integer).

- [Optional Reading] When σ is unknown, we use the sample standard deviation, s, instead, but as a result we also need to use a different set of confidence multipliers (t^*) associated with the t distribution. The interval is therefore

  $$\overline{x} \pm t^*\frac{s}{\sqrt{n}}$$

- [Optional Reading] These new multipliers have the added complexity that they depend not only on the level of confidence, but also on the sample size. Software is therefore very useful for calculating confidence intervals in this case.

- [Optional Reading] For large values of n, the t^* multipliers are not that different from the z^* multipliers, and therefore using the interval formula:

  $$\overline{x} \pm z^*\frac{s}{\sqrt{n}}$$

  for μ when σ is unknown provides a pretty good approximation.

10.4 Interval Estimation: Confidence Interval for the Population Proportion [NOT COVERED]

11 Module 11 | Hypothesis Testing

Learning Objectives

- Explain the logic behind and the process of hypotheses testing. In particular, explain what the p-value is and how it is used to draw conclusions.
- In a given context, specify the null and alternative hypotheses for the population proportion and mean.
- Carry out hypothesis testing for the population proportion and mean (when appropriate), and draw conclusions in context.
- Apply the concepts of: sample size, statistical significance vs. practical importance, and the relationship between hypothesis testing and confidence intervals.
- Determine the likelihood of making type I and type II errors, and explain how to reduce them, in context.

We are in the middle of the part of the course that has to do with inference for one variable.

So far, we talked about point estimation and learned how interval estimation enhances it by quantifying the magnitude of the estimation error (with a certain level of confidence) in the form of the margin of error. The result is the confidence interval – an interval that, with a certain confidence, we believe captures the unknown parameter.

We are now moving to the other kind of inference, *hypothesis testing*. We say that hypothesis testing is "the other kind" because, unlike the inferential methods we presented so far, where the goal was *estimating* the unknown parameter, the idea, logic and goal of hypothesis testing are quite different.

In the first part of this module we will discuss the idea behind hypothesis testing, explain how it works, and introduce new terminology that emerges in this form of inference. The next two parts will be more specific and will discuss hypothesis testing for the population proportion (p), and for the population mean (μ).

11.1 Overview

11.1.1 Module 11 | Overview (1 of 4)

Learning Objectives

- Explain the logic behind and the process of hypotheses testing. In particular, explain what the p-value is and how it is used to draw conclusions.

The purpose of this section is to gradually build your understanding about how statistical hypothesis testing works. We start by explaining the general logic behind the process of hypothesis testing. Once we are confident that you understand this logic, we will add some more details and terminology.

General Idea and Logic of Hypothesis Testing To start our discussion about the idea behind statistical hypothesis testing, consider the following example:

> ***Example* (P) 1**
>
> A case of suspected cheating on an exam is brought in front of the disciplinary committee at a certain university.
>
> There are *two* opposing *claims* in this case:
>
> - The *student's claim:* I did not cheat on the exam.
> - The *instructor's claim:* The student did cheat on the exam.
>
> Adhering to the principle *"innocent until proven guilty,"* the committee asks the instructor for *evidence* to support his claim. The instructor explains that the exam had two versions, and shows the committee members that on three separate exam questions, the student used in his solution numbers that were given in the other version of the exam.
>
> The committee members all agree that *it would be extremely unlikely to get evidence like that if the student's claim of not cheating had been true*. In other words, the committee members all agree that the instructor brought forward strong enough evidence to reject the student's claim, and conclude that the student did cheat on the exam.

What does this example have to do with statistics?

While it is true that this story seems unrelated to statistics, it captures all the elements of hypothesis testing and the logic behind it. Before you read on to understand why, it would be useful to read the example again. Please do so now.

Statistical hypothesis testing is defined as:

Assessing evidence provided by the data in favor of or against some claim about the population.

Here is how the process of statistical hypothesis testing works:

1. We have *two claims* about what is going on in the population. Let's call them for now *claim 1* and *claim 2*. Much like the story above, where the student's claim is challenged by the instructor's claim, claim 1 is challenged by claim 2.

(*Comment:* as you'll see in the examples that follow, these claims are usually about the value of population parameter(s) or about the existence or nonexistence of a relationship between two variables in the population).

2. We choose a sample, collect relevant data and summarize them (this is similar to the instructor collecting evidence from the student's exam).
3. We figure out how likely it is to observe data like the data we got, had claim 1 been true. (Note that the wording "how likely ..." implies that this step requires some kind of probability calculation). In the story, the committee members assessed how likely it is to observe the evidence like that which the instructor provided, had the student's claim of not cheating been true.
4. Based on what we found in the previous step, we make our decision:
   - If we find that if claim 1 were true it would be extremely unlikely to observe the data that we observed, then we have strong evidence against claim 1, and we reject it in favor of claim 2.
   - If we find that if claim 1 were true observing the data that we observed is not very unlikely, then we do not have enough evidence against claim 1, and therefore we cannot reject it in favor of claim 2.

In our story, the committee decided that it would be extremely unlikely to find the evidence that the instructor provided had the student's claim of not cheating been true. In other words, the members felt that it is extremely unlikely that it is just a coincidence that the student used the numbers from the other version of the exam on three separate problems. The committee members therefore decided to reject the student's claim and concluded that the student had, indeed, cheated on the exam. (Wouldn't you conclude the same?)

Hopefully this example helped you understand the logic behind hypothesis testing. To strengthen your understanding of the process of hypothesis testing and the logic behind it, let's look at three statistical examples.

***Example* (Q) 1**

A recent study estimated that 20% of all college students in the United States smoke. The head of Health Services at Goodheart University suspects that the proportion of smokers may be lower there. In hopes of confirming her claim, the head of Health Services chooses a random sample of 400 Goodheart students, and finds that 70 of them are smokers.

Let's analyze this example using the 4 steps outlined above:

1. *Stating the claims:*

   There are two claims here:

   - *claim 1:* The proportion of smokers at Goodheart is 0.20.
   - *claim 2:* The proportion of smokers at Goodheart is less than 0.20.

   Claim 1 basically says "nothing special goes on in Goodheart University; the proportion of smokers there is no different from the proportion in the entire country." This claim is challenged by the head of Health Services, who suspects that the proportion of smokers at Goodheart is lower.

2. *Choosing a sample and collecting data:*

   A sample of $n = 400$ was chosen, and summarizing the data revealed that the sample proportion of smokers is $\hat{p} = \frac{70}{400} = 0.175$.

   While it is true that 0.175 is less than 0.20, it is not clear whether this is strong enough evidence against claim 1.

3. *Assessment of evidence:*

   In order to assess whether the data provide strong enough evidence against claim 1, we need to ask ourselves: How surprising is it to get a sample proportion as low as $\hat{p} = 0.175$ (or lower), assuming claim 1 is true?

   In other words, we need to find how likely it is that in a random sample of size $n = 400$ taken from a population where the proportion of smokers is $p = 0.20$ we'll get a sample proportion as low as $\hat{p} = 0.175$ (or lower).

   It turns out that the probability that we'll get a sample proportion as low as $\hat{p} = 0.175$ (or lower) in such a sample is roughly 0.106 (do not worry about how this was calculated at this point).

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4. *Conclusion:*

   Well, we found that if claim 1 were true there is a probability of 0.106 of observing data like that observed.

   Now you have to decide ...

   Do you think that a probability of 0.106 makes our data rare enough (surprising enough) under claim 1 so that the fact that we *did* observe it is enough evidence to reject claim 1?

   Or do you feel that a probability of 0.106 means that data like we observed are not very likely when claim 1 is true, but they are not unlikely enough to conclude that getting such data is sufficient evidence to reject claim 1.

   Basically, this is your decision. However, it would be nice to have some kind of guideline about what is generally considered surprising enough.

11.1.2 Module 11 | Overview (2 of 4)

Learning Objectives

- Explain the logic behind and the process of hypotheses testing. In particular, explain what the p-value is and how it is used to draw conclusions.

Example **(R) 2**

A certain prescription allergy medicine is supposed to contain an average of 245 parts per million (ppm) of a certain chemical. If the concentration is higher than 245 ppm, the drug will likely cause unpleasant side effects, and if the concentration is below 245 ppm, the drug may be ineffective. The manufacturer wants to check whether the mean concentration in a large shipment is the required 245 ppm or not. To this end, a random sample of 64 portions from the large shipment is tested, and it is found that the sample mean concentration is 250 ppm with a sample standard deviation of 12 ppm. Let's analyze this example according to the four steps of hypotheses testing we outlined on the previous page:

1. *Stating the claims:*

   - *Claim 1:* The mean concentration in the shipment is the required 245 ppm.
   - *Claim 2:* The mean concentration in the shipment is not the required 245 ppm.

   Note that again, claim 1 basically says: "There is nothing unusual about this shipment, the mean concentration is the required 245 ppm." This claim is challenged by the manufacturer, who wants to check whether that is, indeed, the case or not.

2. *Choosing a sample and collecting data:*

   A sample of $n = 64$ portions is chosen and after summarizing the data it is found that the sample concentration is $\bar{x} = 250$ and the sample standard deviation is $s = 12$.

   Is the fact that $\bar{x} = 250$ is different from 245 strong enough evidence to reject claim 1 and conclude that the mean concentration in the whole shipment is not the required 245? In other words, do the data provide strong enough evidence to reject claim 1?

3. *Assessing the evidence:*

   In order to assess whether the data provide strong enough evidence against claim 1, we need to ask ourselves the following question: If the mean concentration in the whole shipment were really the required 245 ppm (i.e., if claim 1 were true), how surprising would it be to observe a sample of 64 portions where the sample mean concentration is off by 5 ppm or more (as we did)? It turns out that it would be extremely unlikely to get such a result if the mean concentration were really the required 245. There is only a probability of 0.0007 (i.e., 7 in 10,000) of that happening. (Do not worry about how this was calculated at this point.)

4. *Making conclusions:*

   Here, it is pretty clear that a sample like the one we observed is extremely rare (or extremely unlikely) if the mean concentration in the shipment were really the required 245 ppm. The fact that we *did* observe such a sample therefore provides strong evidence against claim 1, so we reject it and conclude with very little doubt that the mean concentration in the shipment is not the required 245 ppm.

Do you think that you're getting it? Let's make sure, and look at another example.

Example (S) 3

Is there a relationship between gender and combined scores (Math + Verbal) on the SAT exam?

Following a report on the College Board website, which showed that in 2003, males scored generally higher than females on the SAT exam, an educational researcher wanted to check whether this was also the case in her school district. The researcher chose random samples of 150 males and 150 females from her school district, collected data on their SAT performance and found the following:

Males			Females		
n	mean	standard deviation	n	mean	standard deviation
150	1025	212	150	1010	206

Again, let's see how the process of hypothesis testing works for this example:

1. *Stating the claims:*

    - *Claim 1:* Performance on the SAT is not related to gender (males and females score the same).
    - *Claim 2:* Performance on the SAT is related to gender – males score higher.

    Note that again, claim 1 basically says: "There is nothing going on between the variables SAT and gender." Claim 2 represents what the researcher wants to check, or suspects might actually be the case.

2. *Choosing a sample and collecting data:*

    Data were collected and summarized as given above.

    Is the fact that the sample mean score of males (1,025) is higher than the sample mean score of females (1,010) by 15 points strong enough information to reject claim 1 and conclude that in this researcher's school district, males score higher on the SAT than females?

3. *Assessment of evidence:*

    In order to assess whether the data provide strong enough evidence against claim 1, we need to ask ourselves: If SAT scores are in fact not related to gender (claim 1 is true), how likely is it to get data like the data we observed, in which the difference between the males' average and females' average score is as high as 15 points or higher? It turns out that the probability of observing such a sample result if SAT score is not related to gender is approximately 0.29 (Again, do not worry about how this was calculated at this point).

4. *Conclusion:*

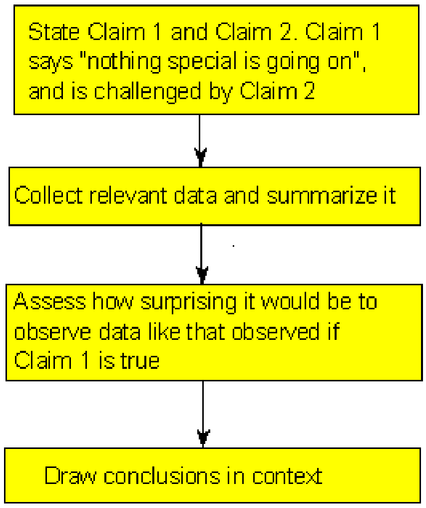
    Here, we have an example where observing a sample like the one we observed is definitely not surprising (roughly 30% chance) if claim 1 were true (i.e., if indeed there is no difference in SAT scores between males and females). We therefore conclude that our data does not provide enough evidence for rejecting claim 1.

Comment Go back and read the conclusion sections of the three examples, and pay attention to the wording. Note that there are two type of conclusions:

- "The data provide enough evidence to reject claim 1 and accept claim 2"; or
- "The data do not provide enough evidence to reject claim 1."

In particular, note that in the second type of conclusion *we did not say:* "*I accept claim 1*," but only "*I don't have enough evidence to reject claim 1*." We will come back to this issue later, but this is a good place to make you aware of this subtle difference.

Hopefully by now, you understand the logic behind the statistical hypothesis testing process. Here is a summary:



11.1.3 Module 11 | Overview (3 of 4)

Learning Objectives

- Explain the logic behind and the process of hypotheses testing. In particular, explain what the *p*-value is and how it is used to draw conclusions.

More Details and Terminology Now that we understand the general idea of how statistical hypothesis testing works, let's go back to each of the steps and delve slightly deeper, getting more details and learning some terminology.

Hypothesis testing step 1: Stating the claims.

In all three examples, our aim is to decide between two opposing points of view, Claim 1 and Claim 2. In hypothesis testing, *Claim 1* is called the *null hypothesis* (denoted "H_0"), and *Claim 2* plays the role of the *alternative hypothesis* (denoted "H_a"). As we saw in the three examples, the null hypothesis suggests nothing special is going on; in other words, there is no change from the status quo, no difference from the traditional state of affairs, no relationship. In contrast, the alternative hypothesis disagrees with this, stating that something is going on, or there is a change from the status quo, or there is a difference from the traditional state of affairs. The alternative hypothesis, H_a, usually represents what we want to check or what we suspect is really going on.

Let's go back to our three examples and apply the new notation:

In example 1:

- H_0: The proportion of smokers at Goodheart is 0.20.
- H_a: The proportion of smokers at Goodheart is less than 0.20.

In example 2:

- H_0: The mean concentration in the shipment is the required 245 ppm.
- H_a: The mean concentration in the shipment is not the required 245 ppm.

In example 3:

- H_0: Performance on the SAT is not related to gender (males and females score the same).
- H_a: Performance on the SAT is related to gender – males score higher.

Hypothesis testing step 2: Choosing a sample and collecting data.

This step is pretty obvious. This is what inference is all about. You look at sampled data in order to draw conclusions about the entire population. In the case of hypothesis testing, based on the data, you draw conclusions about whether or not there is enough evidence to reject H_0.

There is, however, one detail that we would like to add here. In this step we collect data and *summarize* it. Go back and look at the second step in our three examples. Note that in order to summarize the data we used simple sample statistics such as the sample proportion $\hat{p}$, sample mean ($\overline{x}$) and the sample standard deviation (s).

In practice, you go a step further and use these sample statistics to summarize the data with what's called a *test statistic*. We are not going to go into any details right now, but we will discuss test statistics when we go through the specific tests.

Hypothesis testing step 3: Assessing the evidence.

As we saw, this is the step where we calculate how likely is it to get data like that observed when H_0 true. In a sense, this is the heart of the process, since we draw our conclusions based on this probability. If this probability is very small (see example 2), then that means that it would be very surprising to get data like that observed if H_0 were true. The fact that we *did* observe such data is therefore evidence against H_0, and we should reject it. On the other hand, if this probability is not very small (see example 3) this means that observing data like that observed is not very surprising if H_0 were true, so the fact that we observed such data does not provide evidence against H_0. This crucial probability, therefore, has a special name. It is called the p-value of the test.

In our three examples, the p-values were given to you (and you were reassured that you didn't need to worry about how these were derived):

- Example 1: p-value = 0.106
- Example 2: p-value = 0.0007
- Example 3: p-value = 0.29

Obviously, the smaller the p-value, the more surprising it is to get data like ours when H_0 is true, and therefore, the stronger the evidence the data provide against H_0. Looking at the three p-values of our three examples, we see that the data that we observed in example 2 provide the strongest evidence against the null hypothesis, followed by example 1, while the data in example 3 provides the least evidence against H_0.

11.1.4 Comments:

Right now we will not go into specific details about p-value calculations, but just mention that since the p-value is the probability of getting *data* like those observed when H_0 is true, it would make sense that the calculation of the p-value will be based on the data summary, which, as we mentioned, is the test statistic. Indeed, this is the case. In practice, we will mostly use software to provide the p-value for us.

It should be noted that in the past, before statistical software was such an integral part of intro stats courses it was common to use critical values (rather than p-values) in order to assess the evidence provided by the data. While this courses focuses on p-values, we will provide some details about the critical values approach later in this module for those students who are interested in learning more about it.

11.1.5 Module 11 | Overview (4 of 4)

Learning Objectives

- Explain the logic behind and the process of hypotheses testing. In particular, explain what the p-value is and how it is used to draw conclusions.

Hypothesis testing step 4: Making conclusions.

Since our conclusion is based on how small the p-value is, or in other words, how surprising our data are when H_0 is true, it would be nice to have some kind of guideline or cutoff that will help determine how small the p-value must be, or how "rare" (unlikely) our data must be when H_0 is true, for us to conclude that we have enough evidence to reject H_0.

This cutoff exists, and because it is so important, it has a special name. It is called the *significance level of the test* and is usually denoted by the Greek letter α. The most commonly used significance level is $\alpha = 0.05$ (or 5%). This means that:

- if the p-value < α (usually 0.05), then the data we got is considered to be "rare (or surprising) enough" when H_0 is true, and we say that the data provide significant evidence against H_0, so we reject H_0 and accept H_a.
- if the p-value > α (usually 0.05), then our data are not considered to be "surprising enough" when H_0 is true, and we say that our data do not provide enough evidence to reject H_0 (or, equivalently, that the data do not provide enough evidence to accept H_a).

Important comment about wording.

Another common wording (mostly in scientific journals) is:

"The results are statistically significant" – when the p-value < α.

"The results are not statistically significant" – when the p-value > α.

Comments

1. Although the significance level provides a good guideline for drawing our conclusions, it should not be treated as an incontrovertible truth. There is a lot of room for personal interpretation. What if your p-value is 0.052? You might want to stick to the rules and say ".052 > 0.05 and therefore I don't have enough evidence to reject H_0", but you might decide that 0.052 is small enough for you to believe that H_0 should be rejected.

   It should be noted that scientific journals do consider 0.05 to be the cutoff point for which any p-value below the cutoff indicates enough evidence against H_0, and any p-value above it, *or even equal to it*, indicates there is not enough evidence against H_0.

2. It is important to draw your conclusions *in context*. It is *never enough* to say: "*p-value = ..., and therefore I have enough evidence to reject H_0 at the 0.05 significance level.*"You *should always add:* "... and conclude that ... (what it means in the context of the problem)".

3. Let's go back to the issue of the nature of the two types of conclusions that I can make.

   Either I reject H_0 and accept H_a (when the p-value is smaller than the significance level) or I cannot reject H_0 (when the p-value is larger than the significance level).

As we mentioned earlier, note that the second conclusion does not imply that I accept H_0, but just that I don't have enough evidence to reject it. Saying (by mistake) "I don't have enough evidence to reject H_0 so I accept it" indicates that the data provide evidence that H_0 is true, which is *not necessarily the case*. Consider the following slightly artificial yet effective example:

> ***Example*** (T)
>
> An employer claims to subscribe to an "equal opportunity" policy, not hiring men any more often than women for managerial positions. Is this credible? You're not sure, so you want to test the following *two hypotheses:*
>
> - *H_0:* The proportion of male managers hired is 0.5
> - *H_a:* The proportion of male managers hired is more than 0.5
>
> *Data:* You choose at random three of the new managers who were hired in the last 5 years and find that all 3 are men.
>
> *Assessing Evidence:* If the proportion of male managers hired is really 0.5 (H_0 is true), then the probability that the random selection of three managers will yield three males is therefore 0.5 * 0.5 * 0.5 = 0.125. This is the p-value.
>
> *Conclusion:* Using 0.05 as the significance level, you conclude that since the p-value = 0.125 > 0.05, the fact that the three randomly selected mangers were all males is not enough evidence to reject H_0. In other words, you do not have enough evidence to reject the employer's claim of subscribing to an equal opportunity policy.
>
> However, *the data (all three selected are males) definitely does not provide evidence to accept the employer's claim (H_0).*

11.1.6 Module 11 | Overview Wrap Up

Learning Objectives

- Explain the logic behind and the process of hypotheses testing. In particular, explain what the p-value is and how it is used to draw conclusions.

Let's summarize We learned quite a lot about hypothesis testing. We learned the logic behind it, what the key elements are, and what types of conclusions we can and cannot draw in hypothesis testing.
Here is a quick recap: `https://youtu.be/GzkWcsJyPH4`.

11.2 Type I and Type II Errors

11.2.1 Module 11 | Type I and Type II Errors

Learning Objectives

- Determine the likelihood of making type I and type II errors, and explain how to reduce them, in context.

What Can Go Wrong: Two Types of Errors Statistical investigations involve making decisions in the face of uncertainty. So there is always some chance of making a wrong decision. In hypothesis testing, the following decisions can occur:

- If the null hypothesis is true and we do not reject it, it is a *correct decision.*
- If the null hypothesis is false and we reject it, it is a *correct decision.*
- If the null hypothesis is true, but we reject it. This is a *type I* error.
- If the null hypothesis is false, but we fail to reject it. This is a *type II* error.

Type I and type II errors are not caused by mistakes. They are the result of random chance. The *data* provide evidence for a conclusion that is false. It's no one's fault!

Complete the following table by dragging the appropriate answer into each box:

Example **(U) A Courtroom Analogy for Hypothesis Tests**

Defendants standing trial for a crime are considered innocent until evidence shows they are guilty. It is the job of the prosecution to present evidence that shows the defendant is guilty "beyond a reasonable doubt." It is the job of the defense to challenge this evidence and establish a reasonable doubt. The jury weighs the evidence and makes a decision.

When a jury makes a decision, only two verdicts are possible:

- *Guilty:* The jury concludes that there is enough evidence to convict the defendant. The evidence is so strong that there is not a reasonable doubt of the defendant's guilt.
- *Not guilty:* The jury concludes that there is not enough evidence to conclude beyond a reasonable doubt that the person is guilty.

Notice that a verdict of "not guilty" is not a conclusion that the defendant is innocent. This verdict says only that there is not enough evidence to return a guilty verdict.

How is this like a hypothesis test?

The null hypothesis, H_0, in American courtrooms is "the defendant is innocent." The alternative hypothesis, H_a, is "the defendant is guilty." The evidence presented in the case is the data on which the verdict is based. In a courtroom, the defendant is assumed to be innocent until proven guilty. In a hypothesis test, we assume the null hypothesis is true until the dataindicates otherwise.

The two possible verdicts are similar to the two conclusions that are possible in a hypothesis test.

Reject the null hypothesis: When we reject a null hypothesis, we accept the alternative hypothesis. This is equivalent to a guilty verdict in the courtroom. The evidence is strong enough for the jury to reject the initial assumption

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of innocence. In a hypothesis test, the data is strong enough for us to reject the assumption that the null hypothesis is true.

Fail to reject the null hypothesis: When we fail to reject the null hypothesis, we are delivering a “not guilty” verdict. The jury concludes that the evidence is not strong enough to reject the assumption of innocence. So the data is too weak to support a guilty verdict. We conclude the data is not strong enough to reject the null hypothesis. In other words, the data is too weak to accept the alternative hypothesis.

How does the courtroom analogy relate to Type I and Type II errors?

Type I error: The evidence leads the jury to convict an innocent person. By analogy, we reject a true null hypothesis and accept a false alternative hypothesis.

Type II error: The evidence leads the jury to declare a defendant not guilty, when he is in fact guilty. By analogy, we fail to reject a null hypothesis that is false. In other words, we do not accept an alternative hypothesis when it is really true.

It would be nice to know when each of these errors is happening when we make our decision about the null hypothesis, but statistical decisions are based on evidence gathered through sampling, and our sampling evidence will sometimes fool us. As long as we are making a decision, we will never be able to eliminate the potential for these two types of errors. Thus, we need to learn how to adjust to the consequences of making these types of errors.

What is the probability that we will make a Type I error? If the significance level is 5 percent ($\alpha = 0.05$), then 5 percent of the time, we will reject the null hypothesis, even if it is true. Of course we will not know whether the null hypothesis is true. But if it is, the natural variability that we expect in random samples will produce “rare” results 5 percent of the time.

This makes sense, because when we create the sampling distribution, we assume the null hypothesis is true. We look at the variability in random samples selected from the population described by the null hypothesis.

Similarly, if the significance level is 1 percent, then we can expect the sample results to lead us to reject the null hypothesis 1 percent of the time. In other words, about one in 100 data sets would show “rare” results that contrast with the other 99 data sets, leading us to reject the null hypothesis. If the null hypothesis is actually true, then by chance alone, we will reject a true null hypothesis 1 percent of the time. So the probability of a type I error in this case is 1 percent.

In general, the probability of a type I error is α.

What is the probability that we will make a Type II error? As you have just seen, the probability of a type I error is equal to the significance level, α. The probability of a type II error is much more complicated to calculate, but it is inversely related to the probability of making a type I error. Thus, reducing the chance of making a type II error causes an increase in the likelihood of a type I error.

Decreasing the Chance of Type I or Type II Error How can we decrease the chance of a type I or type II error? Well, decreasing the chance of a type I error increases the chance of a type II error, so we must weigh the consequences of these errors before deciding how to proceed.

Recall that the probability of committing a type I error is α. Why is this? Well, when we choose a level of significance (α), we are choosing a benchmark for rejecting the null hypothesis. If the null hypothesis is true, then the probability that we will reject a true null hypothesis is α. So the smaller α is, the smaller the probability of a type I error there will be.

It is more complicated to calculate the probability of a type II error. The best way to reduce the probability of a type II error is to increase the sample size. But once the sample size is set, larger values of α will decrease the probability of a type II error while increasing the probability of a type I error.

General guidelines for choosing a level of significance:

- If the consequences of a type I error are more serious, choose a small level of significance (α).
- If the consequences of a type II error are more serious, choose a larger level of significance (α). But remember that the level of significance is the probability of committing a type I error.
- In general, we choose the largest level of significance that we can tolerate as the chance of making a type I error.

Note: It is not always the case that one type of error is worse than the other.

11.2.2 Module 11 | Wrap up Type I and Type II Errors

It is important to remember to take into consideration the possibility of the occurrence of Type I or Type II error, when drawing conclusions from hypothesis tests. Thus:

- Whenever there is a failure to reject the null hypothesis, it is possible that a Type II error has occurred. Thus, it is concluded from the results of the study, that there are no significant differences, even though, in reality, there are significant differences.
- Whenever the null hypothesis is rejected, it is possible that a Type I error has been committed. That is, it is concluded from the results of the study, that there are significant differences, when, in reality, there are no differences.

However, it is not possible to know when either Type I or Type II error has occurred.

	We reject the H_0 (accept H_a)	**We fail to reject the H_0** (not enough evidence to accept H_a)
H_0 is true	Type I error	Correct Decision
H_0 is false (H_a is true)	Correct Decision	Type II error